

Supplementary Tables and Figures

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Supplementary Table 1A. Sensitivity of each proposed endpoint to distinguish true Shigella diarrhea of any severity (all diarrhea), criteria-based MSD, and score-based MSD.

Country	Endpoint 1: Any Shigella	Endpoint 2A: Shigella Ct<28.8	Endpoint 2B: Shigella Ct<30.4	Endpoint 3: Any Shigella + no other pathogen	Endpoint 4A: Shigella Ct<28.8 + no other pathogen Ct<30	Endpoint 4B: Shigella Ct<30.4 + no other pathogen Ct<30	Endpoint 5: Culture positive
All diarrhea							
Bangladesh	100.0% (100.0, 100.0)	70.3% (69.6, 71.0)	81.5% (80.9, 82.1)	14.9% (14.3, 15.5)	30.7% (30.0, 31.4)	35.5% (34.8, 36.4)	12.0% (11.5, 12.4)
India	100.0% (100.0, 100.0)	71.7% (70.4, 73.0)	82.5% (81.5, 83.4)	24.8% (23.7, 26.0)	41.1% (39.8, 42.4)	47.3% (45.8, 48.6)	29.0% (27.9, 30.2)
Peru	100.0% (100.0, 100.0)	75.5% (74.7, 76.3)	85.1% (84.6, 85.7)	19.5% (18.7, 20.0)	35.8% (35.0, 36.5)	40.3% (39.5, 41.0)	22.6% (21.8, 23.3)
Pakistan	100.0% (100.0, 100.0)	49.0% (47.8, 50.4)	65.0% (63.9, 66.3)	26.7% (25.4, 27.9)	29.3% (28.3, 30.5)	38.9% (38.0, 40.1)	38.3% (37.2, 39.6)
All Sites	100.0% (100.0, 100.0)	68.8% (68.5, 69.3)	80.2% (79.8, 80.6)	19.4% (19.1, 19.8)	33.4% (33.0, 33.9)	39.1% (38.5, 39.7)	21.5% (21.1, 21.9)
Criteria-based MSD							
Bangladesh	100.0% (100.0, 100.0)	74.8% (73.2, 76.6)	84.6% (83.3, 86.3)	15.1% (13.2, 16.9)	33.0% (31.1, 35.6)	37.3% (35.4, 39.7)	14.6% (13.1, 15.9)
India	100.0% (100.0, 100.0)	74.9% (72.9, 76.5)	84.7% (83.5, 86.1)	24.5% (22.3, 27.2)	42.6% (39.7, 45.4)	48.2% (45.7, 50.6)	31.0% (28.8, 33.8)
Peru	100.0% (100.0, 100.0)	78.7% (77.1, 80.0)	87.2% (86.2, 88.3)	19.7% (18.4, 21.3)	37.6% (36.1, 39.2)	41.7% (39.9, 43.3)	25.7% (24.1, 27.3)
Pakistan	100.0% (100.0, 100.0)	52.2% (49.9, 54.5)	67.8% (65.8, 70.0)	26.5% (24.6, 28.4)	31.1% (29.4, 33.2)	40.3% (38.4, 42.3)	39.4% (37.5, 41.3)
All Sites	100.0% (100.0, 100.0)	70.7% (69.9, 71.8)	81.5% (80.5, 82.3)	21.2% (20.4, 22.4)	35.9% (34.9, 36.9)	41.5% (40.7, 42.5)	27.6% (26.7, 28.4)
Score-based MSD							
Bangladesh	100.0% (100.0, 100.0)	75.3% (73.2, 77.2)	84.9% (83.2, 86.5)	14.8% (13.0, 16.2)	32.6% (30.7, 34.8)	36.8% (34.8, 38.8)	14.5% (13.0, 16.1)
India	100.0% (100.0, 100.0)	74.4% (71.9, 77.7)	84.4% (82.4, 86.6)	23.3% (20.6, 26.2)	41.0% (38.1, 44.1)	46.4% (43.4, 49.4)	28.9% (25.9, 32.3)
Peru	100.0% (100.0, 100.0)	79.1% (77.7, 81.1)	87.5% (86.1, 89.0)	19.6% (17.4, 21.5)	37.8% (35.6, 39.7)	41.7% (39.3, 43.7)	26.2% (24.5, 28.8)
Pakistan	100.0% (100.0, 100.0)	51.7% (50.1, 53.7)	67.4% (65.9, 69.0)	26.1% (24.5, 27.8)	30.6% (29.1, 32.1)	39.8% (38.3, 41.3)	38.7% (37.1, 39.9)
All Sites	100.0% (100.0, 100.0)	66.7% (65.8, 67.8)	78.5% (77.5, 79.3)	21.7% (20.8, 22.6)	34.3% (33.3, 35.4)	40.6% (39.6, 41.6)	29.2% (28.2, 30.2)

Values are mean (95% simulation interval) across countries for the main vaccine scenario.

Supplementary Table 1B. Specificity of each proposed endpoint to distinguish true Shigella diarrhea of any severity (all diarrhea), criteria-based MSD, and score-based MSD.

Country	Endpoint 1: Any Shigella	Endpoint 2A: Shigella Ct<28.8	Endpoint 2B: Shigella Ct<30.4	Endpoint 3: Any Shigella + no other pathogen	Endpoint 4A: Shigella Ct<28.8 + no other pathogen Ct<30	Endpoint 4B: Shigella Ct<30.4 + no other pathogen Ct<30	Endpoint 5: Culture positive
All diarrhea							
Bangladesh	96.4% (96.2, 96.6)	98.8% (98.7, 98.9)	98.2% (98.1, 98.3)	99.7% (99.6, 99.7)	99.6% (99.6, 99.7)	99.5% (99.4, 99.5)	99.9% (99.9, 100.0)
India	98.4% (98.2, 98.6)	99.4% (99.3, 99.6)	99.2% (99.0, 99.3)	99.8% (99.7, 99.9)	99.8% (99.7, 99.8)	99.7% (99.6, 99.8)	99.9% (99.9, 100.0)
Peru	95.2% (95.0, 95.3)	98.0% (97.9, 98.1)	97.2% (97.1, 97.3)	99.4% (99.4, 99.5)	99.3% (99.3, 99.4)	99.1% (99.0, 99.1)	99.8% (99.8, 99.9)
Pakistan	99.0% (98.9, 99.1)	99.9% (99.9, 99.9)	99.8% (99.7, 99.8)	99.8% (99.8, 99.9)	99.9% (99.9, 100.0)	99.9% (99.9, 99.9)	99.9% (99.9, 99.9)
All Sites	96.8% (96.7, 96.9)	98.9% (98.8, 98.9)	98.3% (98.3, 98.4)	99.6% (99.6, 99.7)	99.6% (99.6, 99.7)	99.4% (99.4, 99.5)	99.9% (99.9, 99.9)
Criteria-based MSD							
Bangladesh	90.8% (89.5, 91.9)	94.7% (93.8, 95.5)	93.5% (92.7, 94.5)	99.2% (98.8, 99.5)	98.4% (97.8, 98.9)	98.1% (97.4, 98.6)	99.3% (98.9, 99.6)
India	96.7% (95.4, 97.9)	98.3% (97.1, 99.1)	97.8% (96.4, 98.8)	99.6% (99.0, 100.0)	99.4% (98.7, 99.9)	99.1% (98.3, 99.6)	99.5% (98.9, 99.9)
Peru	91.9% (91.4, 92.7)	95.6% (95.1, 96.0)	94.4% (93.9, 95.0)	99.1% (98.9, 99.2)	98.5% (98.2, 98.8)	98.1% (97.9, 98.4)	99.1% (98.9, 99.3)
Pakistan	97.3% (97.1, 97.6)	99.1% (98.9, 99.4)	98.7% (98.5, 99.0)	99.6% (99.4, 99.7)	99.6% (99.5, 99.7)	99.4% (99.3, 99.6)	99.3% (99.1, 99.4)
All Sites	94.3% (94.0, 94.7)	97.1% (96.9, 97.3)	96.3% (96.0, 96.5)	99.3% (99.2, 99.4)	99.0% (98.9, 99.2)	98.7% (98.6, 98.9)	99.2% (99.1, 99.3)
Score-based MSD							
Bangladesh	91.3% (90.4, 92.1)	95.1% (94.4, 95.9)	94.0% (93.3, 94.8)	99.3% (99.0, 99.6)	98.6% (98.2, 99.0)	98.3% (97.9, 98.7)	99.4% (99.1, 99.6)
India	96.8% (96.1, 97.5)	98.3% (97.7, 98.9)	97.9% (97.1, 98.5)	99.6% (99.3, 99.8)	99.4% (99.0, 99.7)	99.2% (98.8, 99.7)	99.5% (99.2, 99.8)
Peru	92.1% (91.6, 92.6)	95.6% (95.3, 96.0)	94.5% (94.1, 95.0)	99.1% (98.9, 99.3)	98.6% (98.3, 98.8)	98.2% (98.0, 98.5)	99.2% (99.0, 99.4)
Pakistan	97.7% (97.5, 97.9)	99.3% (99.2, 99.4)	99.0% (98.8, 99.1)	99.6% (99.5, 99.7)	99.7% (99.6, 99.8)	99.6% (99.5, 99.7)	99.4% (99.3, 99.5)
All Sites	95.1% (94.9, 95.4)	97.6% (97.5, 97.7)	96.9% (96.7, 97.1)	99.4% (99.3, 99.5)	99.2% (99.1, 99.3)	99.0% (98.9, 99.1)	99.3% (99.3, 99.4)

Values are mean (95% simulation interval) across countries for the main vaccine scenario.

Supplementary Table 2. True and observed vaccine efficacy (VE) when estimated using each endpoint, for *Shigella* diarrhea of any severity (all diarrhea), criteria-based MSD, and score-based MSD

Country	True VE	Endpoint 1: Any <i>Shigella</i>	Endpoint 2A: <i>Shigella</i> Ct<28.8	Endpoint 2B: <i>Shigella</i> Ct<30.4	Endpoint 3: Any <i>Shigella</i> + no other pathogen	Endpoint 4A: <i>Shigella</i> Ct<28.8 + no other pathogen Ct<30	Endpoint 4B: <i>Shigella</i> Ct<30.4 + no other pathogen Ct<30	Endpoint 5: Culture positive
All diarrhea								
Bangladesh	40.0%	31.6% (29.3, 34.0)	39.8% (37.4, 42.4)	37.1% (34.6, 39.6)	35.3% (30.7, 40.6)	41.0% (37.3, 44.6)	38.7% (35.6, 41.5)	49.6% (44.4, 54.8)
India	40.0%	36.9% (33.1, 40.3)	42.7% (39.4, 45.9)	40.8% (36.4, 43.8)	38.0% (30.1, 46.5)	43.1% (37.3, 47.2)	41.3% (35.5, 45.8)	47.2% (41.3, 52.2)
Peru	40.0%	26.1% (23.6, 28.9)	35.8% (33.8, 38.7)	32.6% (30.6, 35.3)	30.8% (24.4, 35.8)	37.4% (33.8, 41.6)	34.8% (30.7, 38.8)	46.3% (41.5, 50.4)
Pakistan	40.0%	33.1% (29.3, 36.0)	46.6% (42.5, 50.3)	42.9% (39.8, 46.4)	35.4% (28.5, 41.0)	46.9% (41.7, 51.5)	43.5% (37.9, 47.6)	42.9% (37.5, 47.9)
All Sites	40.0%	30.1% (28.5, 31.6)	38.8% (37.1, 40.9)	36.0% (34.3, 37.7)	34.4% (31.1, 36.8)	40.8% (38.7, 42.8)	38.4% (36.3, 40.1)	46.3% (43.7, 48.6)
Criteria-based MSD								
Bangladesh	50.9%	46.4% (41.9, 51.5)	51.2% (45.6, 57.9)	49.6% (44.0, 55.3)	47.2% (33.0, 59.6)	51.8% (43.1, 60.0)	50.2% (42.2, 58.3)	58.6% (47.8, 68.5)
India	49.7%	49.1% (42.2, 55.1)	52.5% (44.4, 58.2)	51.2% (43.7, 57.1)	48.8% (34.3, 58.8)	51.8% (40.5, 59.7)	50.7% (41.4, 57.9)	55.2% (46.0, 63.8)
Peru	49.4%	40.5% (36.3, 44.0)	46.5% (41.9, 50.4)	44.5% (40.8, 48.1)	43.3% (32.5, 51.0)	47.7% (40.3, 53.3)	46.0% (38.9, 51.4)	53.5% (46.7, 59.8)
Pakistan	57.1%	51.8% (46.6, 56.0)	59.7% (53.0, 64.3)	57.6% (50.5, 63.8)	52.8% (43.6, 60.6)	59.9% (49.4, 67.3)	57.8% (47.2, 64.2)	57.3% (51.0, 63.9)
All Sites	51.7%	45.9% (43.6, 48.7)	51.0% (48.1, 53.6)	49.5% (47.0, 51.9)	48.0% (42.1, 53.1)	52.1% (47.3, 55.4)	50.7% (46.3, 54.3)	55.8% (51.4, 59.8)
Score-based MSD								
Bangladesh	81.0%	70.6% (67.5, 74.1)	75.4% (72.0, 78.8)	73.9% (70.4, 77.6)	75.1% (66.1, 83.4)	77.7% (73.0, 82.4)	76.5% (71.3, 81.1)	80.6% (74.4, 86.1)
India	67.9%	64.6% (58.9, 69.4)	68.1% (62.2, 73.0)	66.9% (61.4, 71.7)	66.3% (56.7, 76.1)	68.8% (61.8, 74.6)	67.7% (61.0, 73.7)	71.1% (62.4, 79.1)
Peru	73.0%	55.2% (51.6, 58.7)	63.1% (60.2, 66.4)	60.7% (57.4, 63.8)	60.7% (53.3, 67.0)	65.9% (61.1, 70.4)	64.0% (59.0, 69.5)	70.9% (65.2, 76.6)
Pakistan	57.0%	49.6% (46.1, 53.1)	59.2% (54.7, 62.6)	56.6% (52.4, 60.4)	52.1% (44.2, 59.1)	60.1% (52.5, 65.9)	57.7% (51.8, 63.6)	57.0% (51.7, 61.3)
All Sites	68.3%	57.9% (56.2, 59.7)	65.8% (63.8, 67.8)	63.5% (61.3, 65.2)	60.0% (56.1, 64.0)	67.0% (63.9, 69.6)	64.9% (62.2, 67.7)	65.2% (62.0, 67.5)

Observed VE is mean (95% simulation interval). Main vaccine scenario.

Supplementary Table 3. Bias in observed VE for Shigella diarrhea of any severity (all diarrhea), criteria-based MSD, and score-based MSD when using different trial endpoints.

Country	Endpoint 1: Any Shigella	Endpoint 2A: Shigella Ct<28.8	Endpoint 2B: Shigella Ct<30.4	Endpoint 3: Any Shigella + no other pathogen	Endpoint 4A: Shigella Ct<28.8 + no other pathogen Ct<30	Endpoint 4B: Shigella Ct<30.4 + no other pathogen Ct<30	Endpoint 5: Culture positive
All diarrhea							
Bangladesh	-8.4% (-10.7, -6.0)	-0.2% (-2.6, 2.4)	-2.9% (-5.4, -0.4)	-4.7% (-9.3, 0.6)	1.0% (-2.7, 4.6)	-1.3% (-4.4, 1.5)	9.6% (4.4, 14.8)
India	-3.1% (-6.9, 0.3)	2.7% (-0.6, 5.9)	0.8% (-3.6, 3.8)	-2.0% (-9.9, 6.5)	3.1% (-2.7, 7.2)	1.3% (-4.5, 5.8)	7.2% (1.3, 12.2)
Peru	-13.9% (-16.4, -11.1)	-4.2% (-6.2, -1.3)	-7.4% (-9.4, -4.7)	-9.2% (-15.6, -4.2)	-2.6% (-6.2, 1.6)	-5.2% (-9.3, -1.2)	6.3% (1.5, 10.4)
Pakistan	-6.9% (-10.7, -4.0)	6.6% (2.5, 10.3)	2.9% (-0.2, 6.4)	-4.6% (-11.5, 1.0)	6.9% (1.7, 11.5)	3.5% (-2.1, 7.6)	2.9% (-2.5, 7.9)
All Sites	-9.9% (-11.5, -8.4)	-1.2% (-2.9, 0.9)	-4.0% (-5.7, -2.3)	-5.6% (-8.9, -3.2)	0.8% (-1.3, 2.8)	-1.6% (-3.7, 0.1)	6.3% (3.7, 8.6)
Criteria-based MSD							
Bangladesh	-4.5% (-9.0, 0.7)	0.3% (-5.3, 7.1)	-1.3% (-6.9, 4.5)	-3.7% (-17.9, 8.8)	0.9% (-7.7, 9.1)	-0.6% (-8.6, 7.4)	7.8% (-3.0, 17.7)
India	-0.6% (-7.5, 5.3)	2.7% (-5.3, 8.4)	1.5% (-6.1, 7.4)	-1.0% (-15.5, 9.1)	2.1% (-9.2, 10.0)	1.0% (-8.3, 8.2)	5.5% (-3.8, 14.0)
Peru	-8.9% (-13.1, -5.4)	-2.9% (-7.5, 1.0)	-4.9% (-8.6, -1.3)	-6.1% (-16.9, 1.6)	-1.7% (-9.0, 3.9)	-3.4% (-10.5, 2.0)	4.1% (-2.7, 10.4)
Pakistan	-5.3% (-10.5, -1.1)	2.6% (-4.0, 7.2)	0.6% (-6.6, 6.8)	-4.2% (-13.4, 3.6)	2.8% (-7.6, 10.3)	0.8% (-9.8, 7.2)	0.3% (-6.1, 6.8)
All Sites	-5.7% (-8.1, -3.0)	-0.7% (-3.6, 1.9)	-2.2% (-4.7, 0.2)	-3.7% (-9.6, 1.4)	0.4% (-4.4, 3.7)	-1.0% (-5.4, 2.6)	4.1% (-0.3, 8.1)
Score-based MSD							
Bangladesh	-10.4% (-13.5, -7.0)	-5.6% (-9.0, -2.2)	-7.1% (-10.6, -3.5)	-5.9% (-15.0, 2.4)	-3.3% (-8.0, 1.4)	-4.5% (-9.7, 0.1)	-0.4% (-6.7, 5.1)
India	-3.2% (-9.0, 1.5)	0.2% (-5.7, 5.1)	-1.0% (-6.5, 3.8)	-1.6% (-11.2, 8.2)	0.9% (-6.0, 6.7)	-0.2% (-6.8, 5.8)	3.2% (-5.5, 11.2)
Peru	-17.8% (-21.3, -14.3)	-9.8% (-12.8, -6.6)	-12.3% (-15.6, -9.2)	-12.3% (-19.6, -5.9)	-7.1% (-11.9, -2.6)	-9.0% (-14.0, -3.5)	-2.1% (-7.8, 3.6)
Pakistan	-7.4% (-10.9, -3.9)	2.1% (-2.3, 5.6)	-0.4% (-4.6, 3.4)	-4.9% (-12.9, 2.1)	3.1% (-4.5, 8.9)	0.7% (-5.3, 6.6)	-0.0% (-5.3, 4.3)
All Sites	-10.4% (-12.1, -8.6)	-2.5% (-4.5, -0.5)	-4.8% (-7.1, -3.1)	-8.3% (-12.2, -4.3)	-1.3% (-4.4, 1.3)	-3.4% (-6.1, -0.7)	-3.1% (-6.3, -0.8)

Bias is measured as the absolute difference between observed and true VE. Values are mean (95% simulation interval). Main vaccine scenario.

Supplementary Table 4A. Sensitivity across all sites comparing the main vaccine scenario to varying each vaccine effect to higher and lower values than those in the main scenario.

Severity	Vaccine Scenario	Endpoint 1: Any Shigella	Endpoint 2A: Shigella Ct<28.8	Endpoint 2B: Shigella Ct<30.4	Endpoint 3: Any Shigella + no other pathogen	Endpoint 4A: Shigella Ct<28.8 + no other pathogen Ct<30	Endpoint 4B: Shigella Ct<30.4 + no other pathogen Ct<30	Endpoint 5: Culture positive
All diarrhea	Main	100.0%	68.8%	80.2%	19.4%	33.4%	39.1%	21.5%
	VE against infection							
	0%	100.0% (+0.0%)	68.8% (-0.0%)	80.2% (-0.0%)	19.4% (+0.0%)	33.4% (-0.0%)	39.1% (-0.0%)	21.5% (+0.0%)
	20%	100.0% (+0.0%)	68.8% (+0.0%)	80.2% (-0.0%)	19.4% (+0.0%)	33.4% (-0.0%)	39.1% (-0.0%)	21.5% (+0.0%)
	VE against disease							
	30%	100.0% (+0.0%)	68.6% (-0.3%)	80.0% (-0.2%)	19.4% (+0.0%)	33.3% (-0.1%)	39.0% (-0.1%)	21.3% (-0.1%)
	50%	100.0% (+0.0%)	69.1% (+0.3%)	80.4% (+0.2%)	19.5% (+0.0%)	33.5% (+0.1%)	39.2% (+0.1%)	21.6% (+0.1%)
	Target VE against MSD							
	40%	100.0% (+0.0%)	69.0% (+0.2%)	80.3% (+0.1%)	19.4% (+0.0%)	33.5% (+0.1%)	39.1% (+0.1%)	21.6% (+0.1%)
	80%	100.0% (+0.0%)	68.6% (-0.2%)	80.0% (-0.2%)	19.4% (+0.0%)	33.3% (-0.1%)	39.0% (-0.1%)	21.4% (-0.1%)
Criteria- based MSD	Reduction in pathogen quantity							
	0%	100.0% (+0.0%)	70.6% (+1.8%)	81.5% (+1.3%)	19.4% (+0.0%)	34.3% (+0.9%)	39.7% (+0.6%)	22.6% (+1.1%)
	20%	100.0% (+0.0%)	66.7% (-2.2%)	78.6% (-1.7%)	19.4% (+0.0%)	32.4% (-1.1%)	38.3% (-0.8%)	20.4% (-1.0%)
	Main	100.0%	70.7%	81.5%	21.2%	35.9%	41.5%	27.6%
	VE against infection							
	0%	100.0% (+0.0%)	70.7% (+0.0%)	81.5% (+0.0%)	21.2% (+0.0%)	35.9% (+0.0%)	41.6% (+0.0%)	27.6% (-0.0%)
	20%	100.0% (+0.0%)	70.7% (-0.0%)	81.5% (+0.0%)	21.2% (+0.0%)	35.9% (-0.0%)	41.5% (-0.0%)	27.6% (+0.0%)
	VE against disease							
	30%	100.0% (+0.0%)	70.7% (-0.0%)	81.5% (+0.0%)	21.2% (-0.0%)	35.8% (-0.1%)	41.5% (-0.1%)	27.5% (-0.1%)
	50%	100.0% (+0.0%)	70.6% (-0.1%)	81.4% (-0.1%)	21.2% (+0.0%)	35.8% (-0.1%)	41.5% (-0.0%)	27.7% (+0.1%)
Score- based MSD	Target VE against MSD							
	40%	100.0% (+0.0%)	70.2% (-0.5%)	81.1% (-0.4%)	21.3% (+0.0%)	35.6% (-0.3%)	41.4% (-0.2%)	27.5% (-0.1%)
	80%	100.0% (+0.0%)	71.1% (+0.4%)	81.7% (+0.3%)	21.2% (-0.0%)	36.1% (+0.2%)	41.6% (+0.1%)	27.7% (+0.1%)
	Reduction in pathogen quantity							
	0%	100.0% (+0.0%)	72.1% (+1.4%)	82.5% (+1.1%)	21.2% (+0.0%)	36.6% (+0.7%)	42.1% (+0.5%)	28.7% (+1.1%)
	20%	100.0% (+0.0%)	68.9% (-1.8%)	80.1% (-1.3%)	21.2% (+0.0%)	35.0% (-0.9%)	40.8% (-0.7%)	26.5% (-1.1%)
	Main	100.0%	66.7%	78.5%	21.7%	34.3%	40.6%	29.2%
	VE against infection							
	0%	100.0% (+0.0%)	66.7% (+0.0%)	78.5% (+0.0%)	21.7% (+0.0%)	34.3% (+0.0%)	40.6% (+0.0%)	29.2% (-0.0%)
	20%	100.0% (+0.0%)	66.7% (-0.0%)	78.5% (-0.0%)	21.7% (+0.0%)	34.2% (-0.0%)	40.5% (-0.0%)	29.1% (-0.0%)
Score- based MSD	VE against disease							
	30%	100.0% (+0.0%)	66.8% (+0.1%)	78.5% (+0.0%)	21.8% (+0.1%)	34.4% (+0.1%)	40.7% (+0.1%)	29.6% (+0.4%)
	50%	100.0% (+0.0%)	67.0% (+0.3%)	78.7% (+0.2%)	21.7% (+0.0%)	34.4% (+0.1%)	40.7% (+0.1%)	29.2% (+0.1%)
	Target VE against MSD							
	40%	100.0% (+0.0%)	66.9% (+0.2%)	78.6% (+0.2%)	21.4% (-0.3%)	34.1% (-0.2%)	40.3% (-0.3%)	28.2% (-1.0%)
	80%	100.0% (+0.0%)	68.7% (+2.0%)	79.9% (+1.5%)	21.4% (-0.3%)	35.1% (+0.8%)	41.1% (+0.5%)	29.3% (+0.1%)
	Reduction in pathogen quantity							
	0%	100.0% (+0.0%)	67.9% (+1.2%)	79.4% (+0.9%)	21.7% (+0.0%)	34.9% (+0.7%)	41.1% (+0.5%)	29.9% (+0.8%)
	20%	100.0% (+0.0%)	65.2% (-1.5%)	77.3% (-1.2%)	21.7% (+0.0%)	33.5% (-0.8%)	39.9% (-0.6%)	28.4% (-0.8%)

Each cell shows the sensitivity for a given scenario and endpoint, and in parentheses the absolute difference in sensitivity compared to the Main scenario for that endpoint.

Supplementary Table 4B. Specificity across all sites comparing the main vaccine scenario to varying each vaccine effect to higher and lower values than those in the main scenario.

Severity	Vaccine Scenario	Endpoint 1: Any Shigella	Endpoint 2A: Shigella Ct<28.8	Endpoint 2B: Shigella Ct<30.4	Endpoint 3: Any Shigella + no other pathogen	Endpoint 4A: Shigella Ct<28.8 + no other pathogen Ct<30	Endpoint 4B: Shigella Ct<30.4 + no other pathogen Ct<30	Endpoint 5: Culture positive
All diarrhea	Main	96.8%	98.9%	98.3%	99.6%	99.6%	99.4%	99.9%
	VE against infection							
	0%	96.6% (-0.2%)	98.8% (-0.1%)	98.2% (-0.1%)	99.6% (-0.0%)	99.6% (-0.0%)	99.4% (-0.0%)	99.9% (-0.0%)
	20%	97.0% (+0.2%)	98.9% (+0.1%)	98.4% (+0.1%)	99.7% (+0.0%)	99.6% (+0.0%)	99.5% (+0.0%)	99.9% (+0.0%)
	VE against disease							
	30%	97.1% (+0.2%)	98.9% (+0.1%)	98.4% (+0.1%)	99.7% (+0.0%)	99.6% (+0.0%)	99.5% (+0.0%)	99.9% (+0.0%)
	50%	96.5% (-0.3%)	98.8% (-0.1%)	98.2% (-0.1%)	99.6% (-0.0%)	99.6% (-0.0%)	99.4% (-0.0%)	99.9% (-0.0%)
	Target VE against MSD							
	40%	96.8% (+0.0%)	98.9% (+0.0%)	98.3% (+0.0%)	99.6% (+0.0%)	99.6% (+0.0%)	99.4% (+0.0%)	99.9% (+0.0%)
	80%	96.8% (+0.0%)	98.9% (+0.0%)	98.3% (+0.0%)	99.6% (+0.0%)	99.6% (+0.0%)	99.4% (+0.0%)	99.9% (+0.0%)
	Reduction in pathogen quantity							
	0%	96.8% (+0.0%)	98.7% (-0.1%)	98.2% (-0.1%)	99.6% (+0.0%)	99.6% (-0.0%)	99.4% (-0.0%)	99.9% (-0.0%)
	20%	96.8% (+0.0%)	99.0% (+0.1%)	98.5% (+0.1%)	99.6% (+0.0%)	99.7% (+0.0%)	99.5% (+0.0%)	99.9% (+0.0%)
Criteria- based MSD	Main	94.3%	97.1%	96.3%	99.3%	99.0%	98.7%	99.2%
	VE against infection							
	0%	94.1% (-0.2%)	97.0% (-0.1%)	96.2% (-0.1%)	99.3% (-0.0%)	99.0% (-0.0%)	98.7% (-0.0%)	99.2% (-0.0%)
	20%	94.5% (+0.2%)	97.1% (+0.1%)	96.4% (+0.1%)	99.3% (+0.0%)	99.0% (+0.0%)	98.8% (+0.0%)	99.2% (+0.0%)
	VE against disease							
	30%	94.3% (+0.0%)	97.0% (-0.1%)	96.2% (-0.0%)	99.3% (+0.0%)	99.0% (-0.0%)	98.7% (-0.0%)	99.2% (-0.0%)
	50%	94.3% (+0.0%)	97.2% (+0.1%)	96.3% (+0.1%)	99.3% (+0.0%)	99.0% (+0.0%)	98.8% (+0.0%)	99.3% (+0.0%)
	Target VE against MSD							
	40%	94.4% (+0.1%)	97.1% (+0.0%)	96.3% (+0.1%)	99.3% (+0.0%)	99.0% (+0.0%)	98.8% (+0.0%)	99.3% (+0.0%)
	80%	94.2% (-0.1%)	97.1% (-0.0%)	96.2% (-0.0%)	99.3% (-0.0%)	99.0% (-0.0%)	98.7% (-0.0%)	99.2% (-0.0%)
	Reduction in pathogen quantity							
	0%	94.3% (+0.0%)	96.9% (-0.2%)	96.1% (-0.1%)	99.3% (+0.0%)	99.0% (-0.1%)	98.7% (-0.0%)	99.2% (-0.0%)
	20%	94.3% (+0.0%)	97.3% (+0.2%)	96.4% (+0.2%)	99.3% (+0.0%)	99.1% (+0.1%)	98.8% (+0.1%)	99.3% (+0.0%)
Score- based MSD	Main	95.1%	97.6%	96.9%	99.4%	99.2%	99.0%	99.3%
	VE against infection							
	0%	95.0% (-0.2%)	97.5% (-0.1%)	96.8% (-0.1%)	99.4% (-0.0%)	99.2% (-0.0%)	98.9% (-0.0%)	99.3% (-0.0%)
	20%	95.3% (+0.2%)	97.6% (+0.0%)	97.0% (+0.1%)	99.4% (+0.0%)	99.2% (+0.0%)	99.0% (+0.0%)	99.3% (+0.0%)
	VE against disease							
	30%	95.0% (-0.1%)	97.5% (-0.1%)	96.8% (-0.1%)	99.4% (-0.0%)	99.2% (-0.0%)	98.9% (-0.0%)	99.3% (-0.1%)
	50%	95.1% (-0.0%)	97.6% (+0.1%)	97.0% (+0.0%)	99.4% (-0.0%)	99.2% (+0.0%)	99.0% (+0.0%)	99.4% (+0.0%)
	Target VE against MSD							
	40%	95.3% (+0.2%)	97.7% (+0.1%)	97.0% (+0.1%)	99.4% (+0.0%)	99.2% (+0.0%)	99.0% (+0.0%)	99.4% (+0.0%)
	80%	94.9% (-0.2%)	97.5% (-0.1%)	96.8% (-0.2%)	99.4% (-0.0%)	99.2% (-0.0%)	98.9% (-0.1%)	99.3% (-0.1%)
	Reduction in pathogen quantity							
	0%	95.1% (+0.0%)	97.5% (-0.1%)	96.8% (-0.1%)	99.4% (+0.0%)	99.1% (-0.0%)	98.9% (-0.0%)	99.3% (-0.0%)
	20%	95.1% (+0.0%)	97.7% (+0.1%)	97.1% (+0.1%)	99.4% (+0.0%)	99.3% (+0.1%)	99.0% (+0.0%)	99.4% (+0.0%)

Each cell shows the specificity for a given scenario and endpoint, and in parentheses the absolute difference in specificity compared to the Main scenario for that endpoint.

Supplementary Table 5. Mean bias in observed VE across all sites, comparing the main vaccine scenario to varying each vaccine effect to higher and lower values than those in the main scenario.

Severity	Vaccine Scenario	Endpoint 1: Any Shigella	Endpoint 2A: Shigella Ct<28.8	Endpoint 2B: Shigella Ct<30.4	Endpoint 3: Any Shigella + no other pathogen	Endpoint 4A: Shigella Ct<28.8 + no other pathogen Ct<30	Endpoint 4B: Shigella Ct<30.4 + no other pathogen Ct<30	Endpoint 5: Culture positive
All diarrhea	Main	-9.9%	-1.2%	-4.0%	-5.6%	0.8%	-1.6%	6.3%
	VE against infection							
	0%	-11.2% (-1.3%)	-1.7% (-0.5%)	-4.7% (-0.7%)	-6.4% (-0.7%)	0.5% (-0.3%)	-2.1% (-0.5%)	6.2% (-0.1%)
	20%	-8.7% (+1.2%)	-0.7% (+0.5%)	-3.2% (+0.7%)	-4.9% (+0.7%)	1.2% (+0.4%)	-1.1% (+0.5%)	6.5% (+0.2%)
	VE against disease							
	30%	-7.5% (+2.5%)	1.1% (+2.3%)	-1.7% (+2.3%)	-4.2% (+1.4%)	2.8% (+2.0%)	0.2% (+1.8%)	8.5% (+2.2%)
	50%	-12.3% (-2.4%)	-3.1% (-1.9%)	-5.9% (-2.0%)	-6.8% (-1.1%)	-0.6% (-1.4%)	-3.0% (-1.4%)	4.6% (-1.7%)
	Target VE against MSD							
	40%	-9.9% (+0.0%)	-1.6% (-0.4%)	-4.2% (-0.3%)	-5.6% (+0.0%)	0.4% (-0.4%)	-1.9% (-0.3%)	5.6% (-0.7%)
	80%	-9.9% (+0.0%)	-0.7% (+0.5%)	-3.6% (+0.4%)	-5.6% (+0.0%)	1.3% (+0.6%)	-1.2% (+0.4%)	7.1% (+0.8%)
Criteria- based MSD	Reduction in pathogen quantity							
	0%	-9.9% (+0.0%)	-6.1% (-4.9%)	-7.2% (-3.3%)	-5.6% (+0.0%)	-4.0% (-4.7%)	-4.7% (-3.1%)	-1.6% (-7.9%)
	20%	-9.9% (+0.0%)	4.6% (+5.8%)	0.1% (+4.1%)	-5.6% (+0.0%)	6.4% (+5.7%)	2.3% (+3.9%)	13.9% (+7.5%)
	Main	-5.7%	-0.7%	-2.2%	-3.7%	0.4%	-1.0%	4.1%
	VE against infection							
	0%	-6.4% (-0.7%)	-0.9% (-0.3%)	-2.6% (-0.4%)	-4.2% (-0.5%)	0.2% (-0.2%)	-1.2% (-0.2%)	4.1% (-0.0%)
	20%	-5.1% (+0.6%)	-0.3% (+0.3%)	-1.8% (+0.4%)	-3.4% (+0.3%)	0.6% (+0.2%)	-0.7% (+0.3%)	4.3% (+0.2%)
	VE against disease							
	30%	-5.0% (+0.7%)	-0.2% (+0.5%)	-1.7% (+0.5%)	-3.2% (+0.5%)	1.0% (+0.7%)	-0.3% (+0.7%)	5.1% (+1.0%)
	50%	-6.3% (-0.5%)	-0.5% (+0.1%)	-2.2% (-0.0%)	-3.8% (-0.1%)	0.7% (+0.3%)	-0.8% (+0.2%)	4.0% (-0.1%)
Score- based MSD	Target VE against MSD							
	40%	-4.6% (+1.2%)	1.5% (+2.2%)	-0.4% (+1.8%)	-3.3% (+0.4%)	2.2% (+1.9%)	0.4% (+1.4%)	5.8% (+1.7%)
	80%	-6.8% (-1.0%)	-2.2% (-1.6%)	-3.5% (-1.3%)	-4.2% (-0.5%)	-0.9% (-1.3%)	-2.0% (-1.0%)	2.9% (-1.2%)
	Reduction in pathogen quantity							
	0%	-5.7% (+0.0%)	-4.2% (-3.5%)	-4.5% (-2.3%)	-3.7% (+0.0%)	-3.1% (-3.4%)	-3.3% (-2.3%)	-1.9% (-6.0%)
	20%	-5.7% (+0.0%)	3.6% (+4.3%)	0.7% (+2.9%)	-3.7% (+0.0%)	4.6% (+4.2%)	1.8% (+2.8%)	9.9% (+5.8%)
	Main	-10.4%	-2.5%	-4.8%	-8.3%	-1.3%	-3.4%	-3.1%
	VE against infection							
	0%	-11.3% (-0.9%)	-2.8% (-0.3%)	-5.3% (-0.4%)	-9.0% (-0.7%)	-1.6% (-0.3%)	-3.8% (-0.4%)	-3.3% (-0.1%)
	20%	-9.5% (+0.9%)	-2.1% (+0.4%)	-4.4% (+0.5%)	-7.9% (+0.4%)	-1.1% (+0.2%)	-3.1% (+0.3%)	-2.9% (+0.2%)
Score- based MSD	VE against disease							
	30%	-12.4% (-2.0%)	-5.2% (-2.7%)	-7.3% (-2.5%)	-9.6% (-1.4%)	-3.4% (-2.1%)	-5.4% (-2.0%)	-6.7% (-3.6%)
	50%	-11.2% (-0.8%)	-3.0% (-0.5%)	-5.4% (-0.5%)	-8.4% (-0.1%)	-1.7% (-0.4%)	-3.7% (-0.3%)	-3.2% (-0.0%)
	Target VE against MSD							
	40%	-6.2% (+4.2%)	1.4% (+3.9%)	-0.8% (+4.0%)	-4.2% (+4.0%)	2.5% (+3.8%)	0.5% (+3.9%)	4.8% (+7.9%)
	80%	-15.4% (-5.0%)	-9.9% (-7.4%)	-11.3% (-6.5%)	-9.5% (-1.2%)	-6.9% (-5.6%)	-7.9% (-4.5%)	-7.1% (-4.0%)
	Reduction in pathogen quantity							
	0%	-10.3% (+0.1%)	-5.6% (-3.1%)	-7.1% (-2.2%)	-8.2% (+0.1%)	-4.3% (-3.0%)	-5.5% (-2.1%)	-7.0% (-3.9%)
	20%	-10.3% (+0.1%)	1.3% (+3.8%)	-2.1% (+2.8%)	-8.2% (+0.1%)	2.5% (+3.7%)	-0.8% (+2.7%)	0.9% (+4.0%)

Bias is measured as the absolute difference between observed and true VE. Each cell shows the bias for a given scenario and endpoint, and in parentheses the absolute difference in bias (or in observed VE) compared to the Main scenario for that endpoint.

Supplementary Table 6. Percent difference in required trial size across all sites relative to the smallest trial size within each vaccine scenario, to detect an average difference in observed Shigella diarrhea risk between vaccine and placebo groups with 80% power.

Severity	Vaccine Scenario	Endpoint 1: Any Shigella	Endpoint 2A: Shigella Ct<28.8	Endpoint 2B: Shigella Ct<30.4	Endpoint 3: Any Shigella + no other pathogen	Endpoint 4A: Shigella Ct<28.8 + no other pathogen Ct<30	Endpoint 4B: Shigella Ct<30.4 + no other pathogen Ct<30	Endpoint 5: Culture positive
All diarrhea	Main	23.3%	0.0%	0.0%	118.3%	51.0%	48.0%	61.8%
	VE against infection							
	0%	22.2%	0.0%	4.9%	119.2%	49.1%	46.2%	57.1%
	20%	19.0%	0.0%	0.0%	115.6%	51.0%	44.9%	61.8%
	VE against disease							
	30%	108.8%	81.8%	84.4%	166.5%	121.2%	120.0%	121.2%
	50%	14.3%	0.0%	0.0%	108.8%	47.1%	51.4%	59.5%
	Target VE against MSD							
	40%	18.2%	0.0%	0.0%	114.9%	49.1%	51.9%	57.1%
	80%	90.9%	71.4%	75.9%	157.1%	110.0%	107.7%	116.3%
Criteria- based MSD	Reduction in pathogen quantity							
	0%	22.2%	33.3%	26.1%	116.7%	73.0%	64.4%	91.9%
	20%	52.6%	0.0%	19.4%	138.5%	52.6%	60.0%	60.0%
	Main	59.3%	61.8%	53.8%	153.9%	112.6%	105.0%	121.6%
	VE against infection							
	0%	51.9%	54.5%	51.9%	152.4%	107.0%	105.9%	121.6%
	20%	51.0%	59.3%	53.8%	152.5%	111.6%	107.3%	121.6%
	VE against disease							
	30%	107.1%	111.9%	108.8%	175.4%	147.5%	144.7%	154.4%
	50%	76.2%	81.8%	79.1%	160.9%	124.6%	121.2%	136.6%
Score- based MSD	Target VE against MSD							
	40%	94.7%	97.4%	94.7%	171.0%	137.0%	136.5%	144.1%
	80%	83.9%	97.1%	94.1%	166.4%	137.9%	130.8%	147.1%
	Reduction in pathogen quantity							
	0%	51.9%	71.0%	59.6%	152.1%	116.7%	109.1%	137.0%
	20%	80.9%	69.8%	75.6%	164.3%	118.8%	120.0%	125.3%
	Main	5.1%	14.6%	14.6%	121.6%	71.2%	61.8%	88.2%
	VE against infection							
	0%	4.9%	9.5%	4.9%	120.0%	66.7%	59.6%	85.7%
	20%	0.0%	14.6%	10.0%	119.1%	64.3%	59.3%	88.2%
Score- based MSD	VE against disease							
	30%	0.0%	20.7%	14.3%	118.8%	66.7%	63.2%	91.7%
	50%	26.7%	37.5%	20.7%	133.3%	84.4%	79.1%	107.1%
	Target VE against MSD							
	40%	109.1%	104.8%	103.6%	174.7%	142.9%	141.6%	151.2%
	80%	0.0%	20.0%	0.0%	112.2%	66.7%	61.5%	97.1%
	Reduction in pathogen quantity							
	0%	0.0%	26.1%	14.0%	117.5%	75.0%	64.4%	96.1%
	20%	35.3%	30.3%	30.3%	139.1%	80.9%	80.9%	101.8%

Percent difference is calculated as $(n - n_{min}) / ((n + n_{min}) / 2)$, where n_{min} is the smallest sample size across endpoints and disease outcomes (all diarrhea, criteria-based MSD, score-based MSD) within each scenario. Trial sizes assume every Shigella diarrhea episode defined by each endpoint is detected.